CHAPTER 11

Matrix sequences and Markov chains

Sequences $U_{n+1} = AU_n + B$, probabilistic graphs, invariant distributions

Objectives. Study a sequence of column matrices defined by $U_{n+1} = AU_n + B$; model a random evolution with two or three states by a probabilistic graph and a transition matrix; compute the distribution after n steps; find an invariant distribution; study convergence in the two-state case.

Prerequisites. Chapters 9 and 10; conditional probability and the law of total probability; arithmetico-geometric sequences; limit of a geometric sequence.

Fluency

Fluency · No calculator, five minutes.

1. Let $u_{n+1} = 0.5u_n + 2$. Find the real number ℓ such that $\ell = 0.5\ell + 2$.
2. Give the limit of 0.8^n as n tends to $+\infty$.
3. In a tree, $P(A) = 0.3$, $P_A(B) = 0.6$, $P_{\overline{A}}(B) = 0.2$. Compute $P(B)$.
4. Solve the system $0.2a = 0.4b$ and $a + b = 1$.
5. Compute $\begin{pmatrix}1 & 0\end{pmatrix}\begin{pmatrix}0.8 & 0.2\\ 0.4 & 0.6\end{pmatrix}$.
6. What is the sum of the entries of each row of the matrix in question 5?